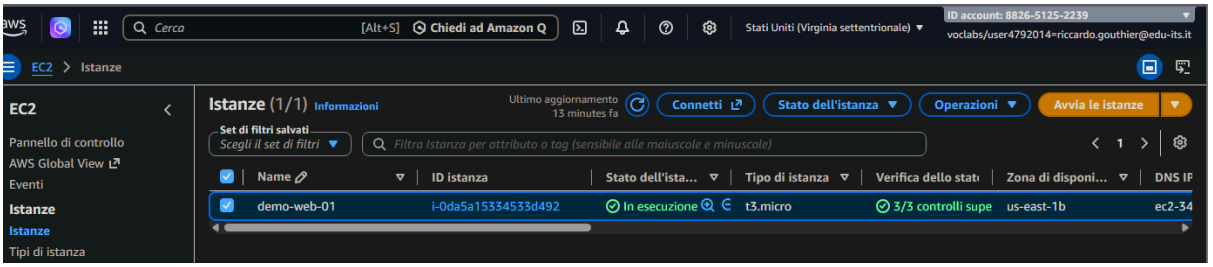
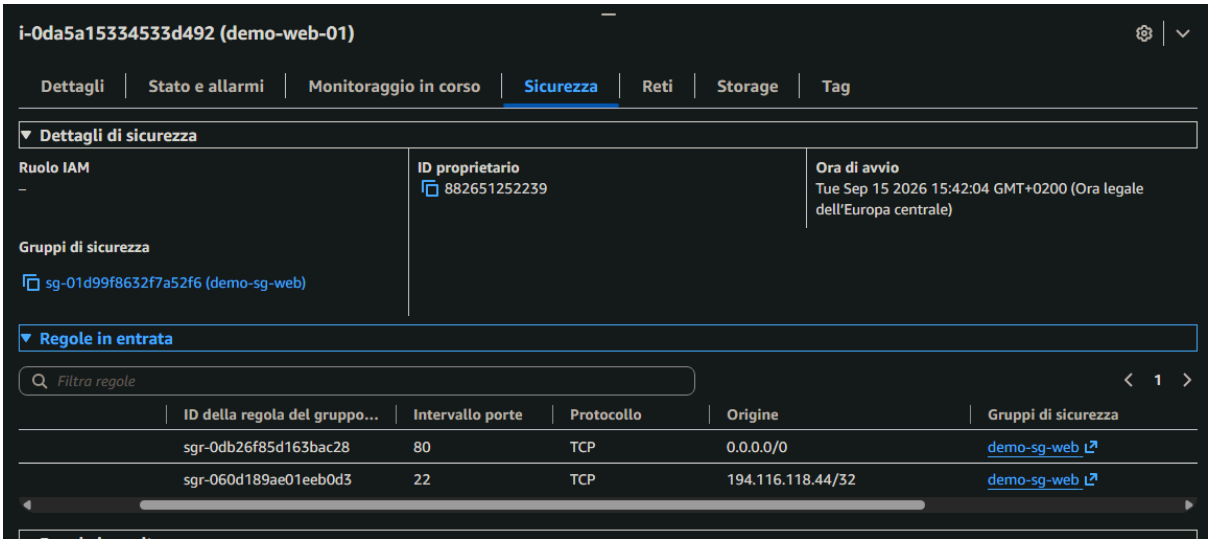
Lab 1

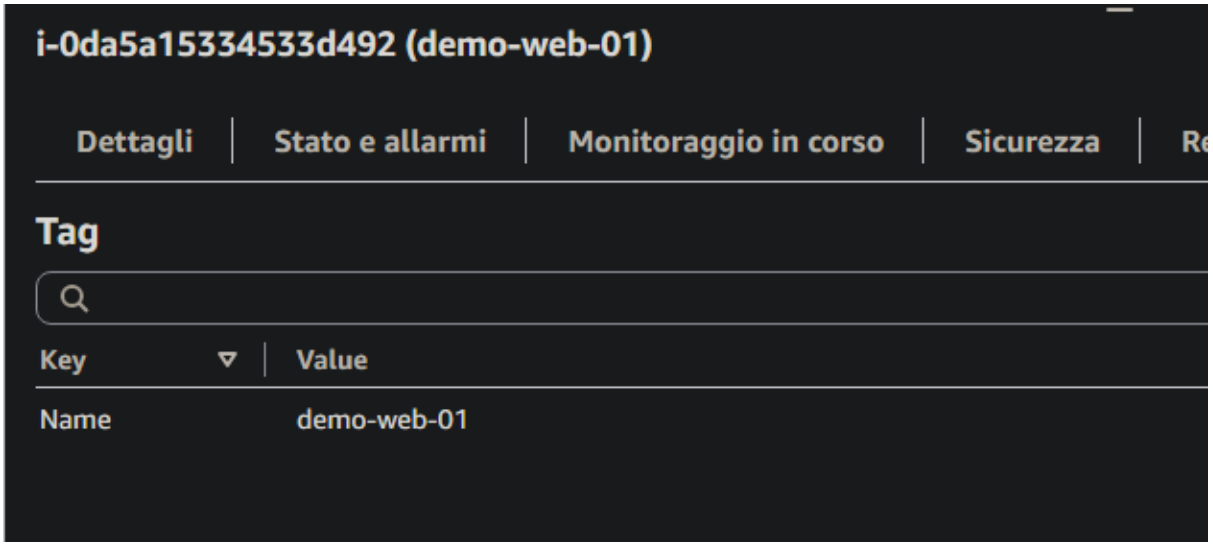
Step 1-3



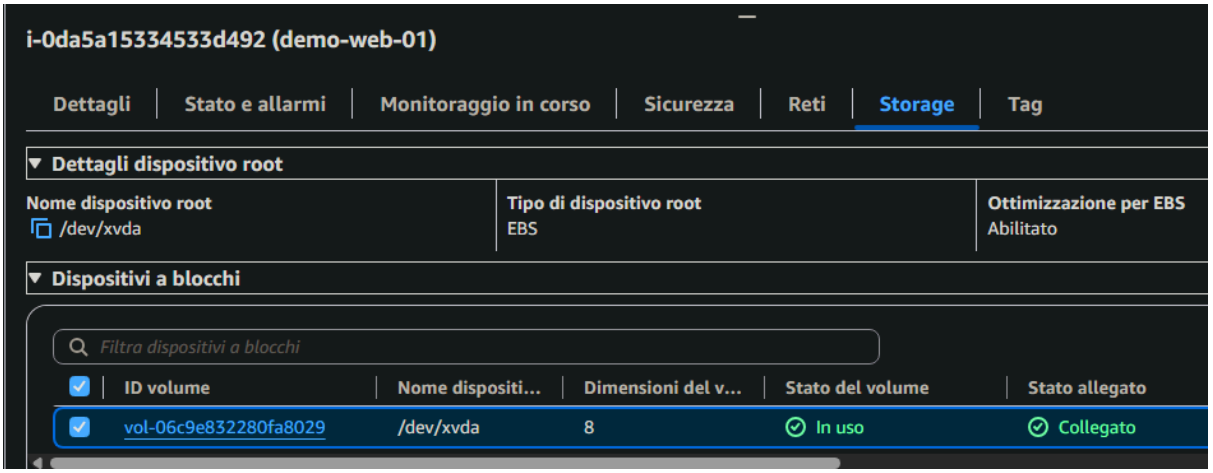
Step 5



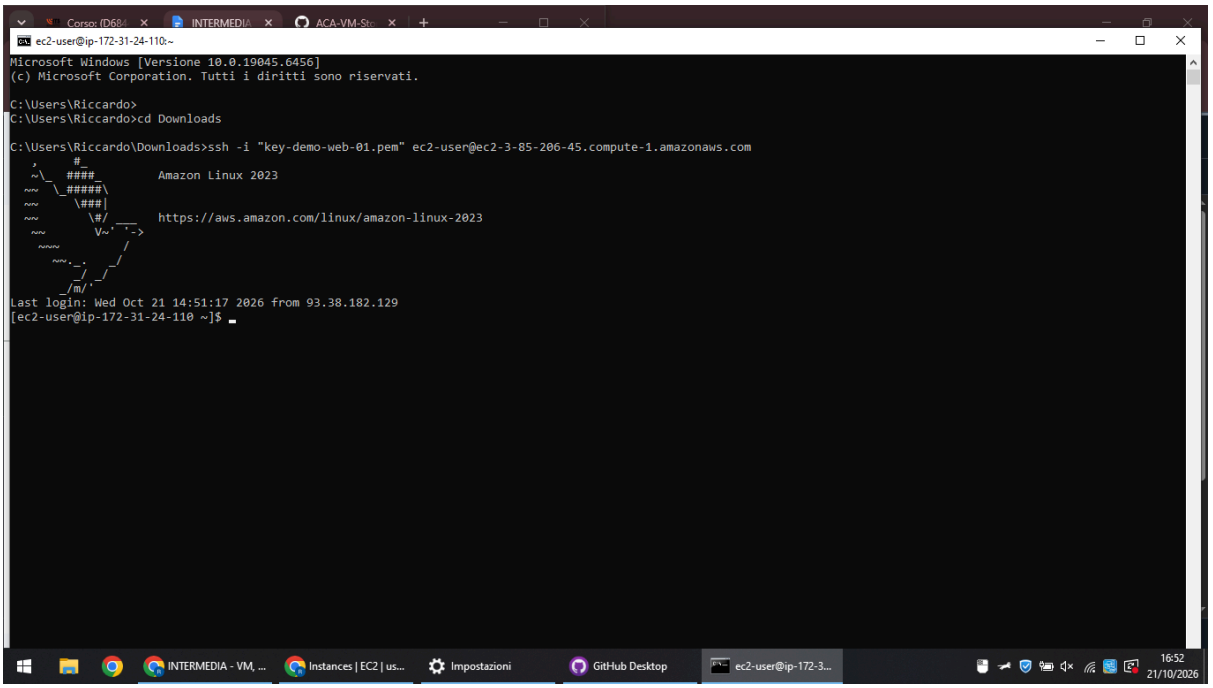
Step 6 Tag



Step 6 Memoria

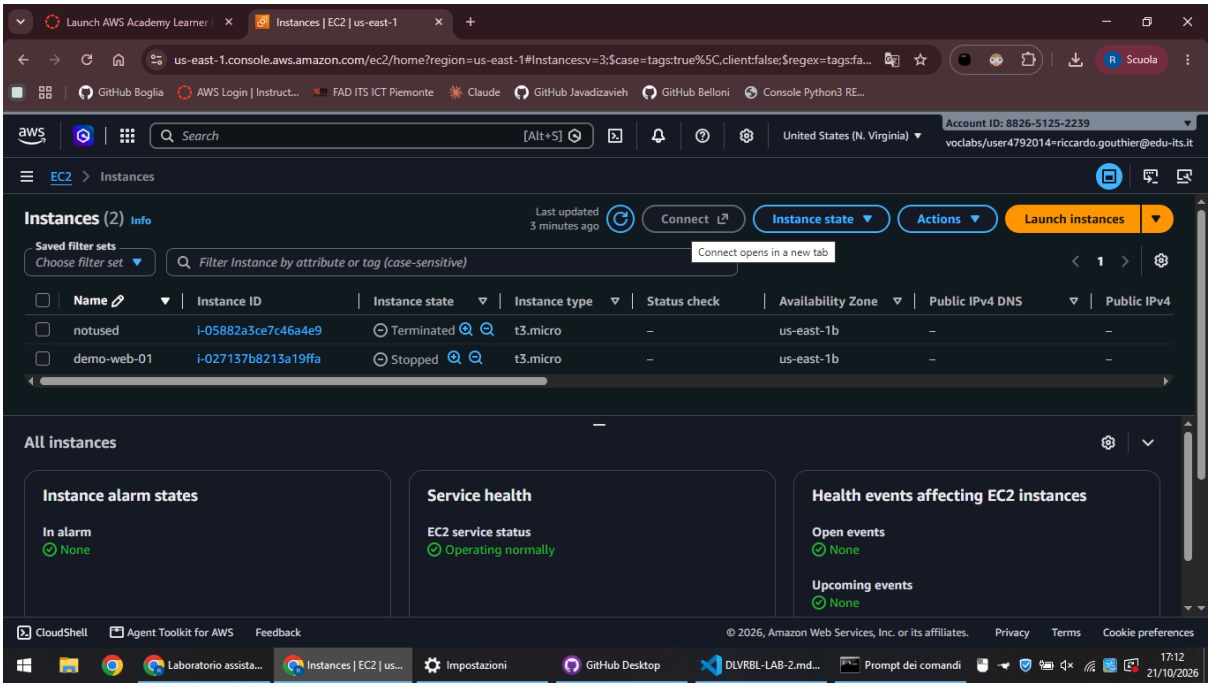


step 7

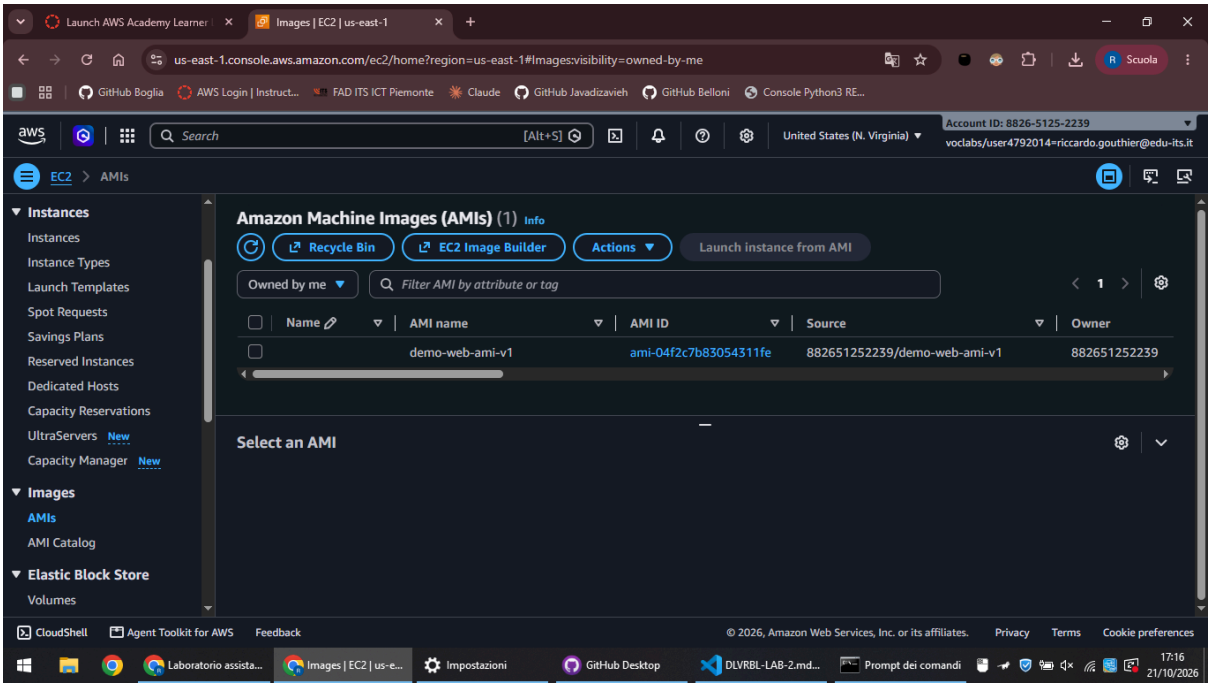


Lab 2

Fermare l'istanza sorgente demo-web-01

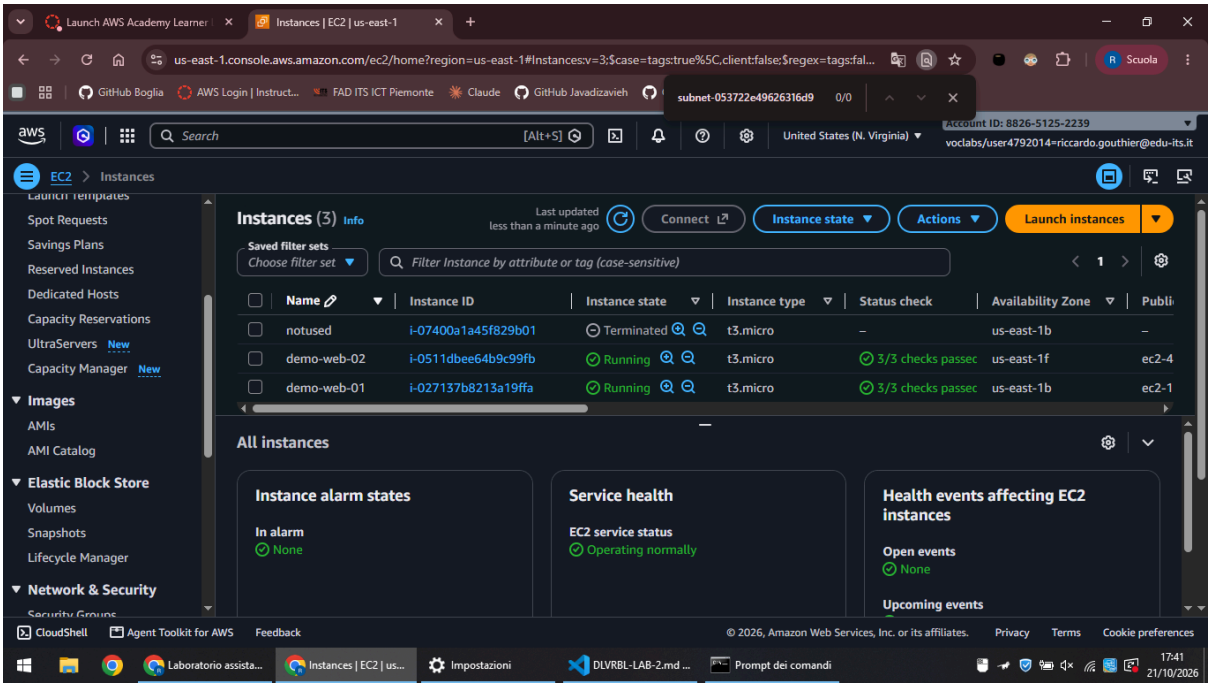


Creare una AMI demo-web-ami-v1



AMI Available

Riavvia demo-web-01 , lancia demo-web-02

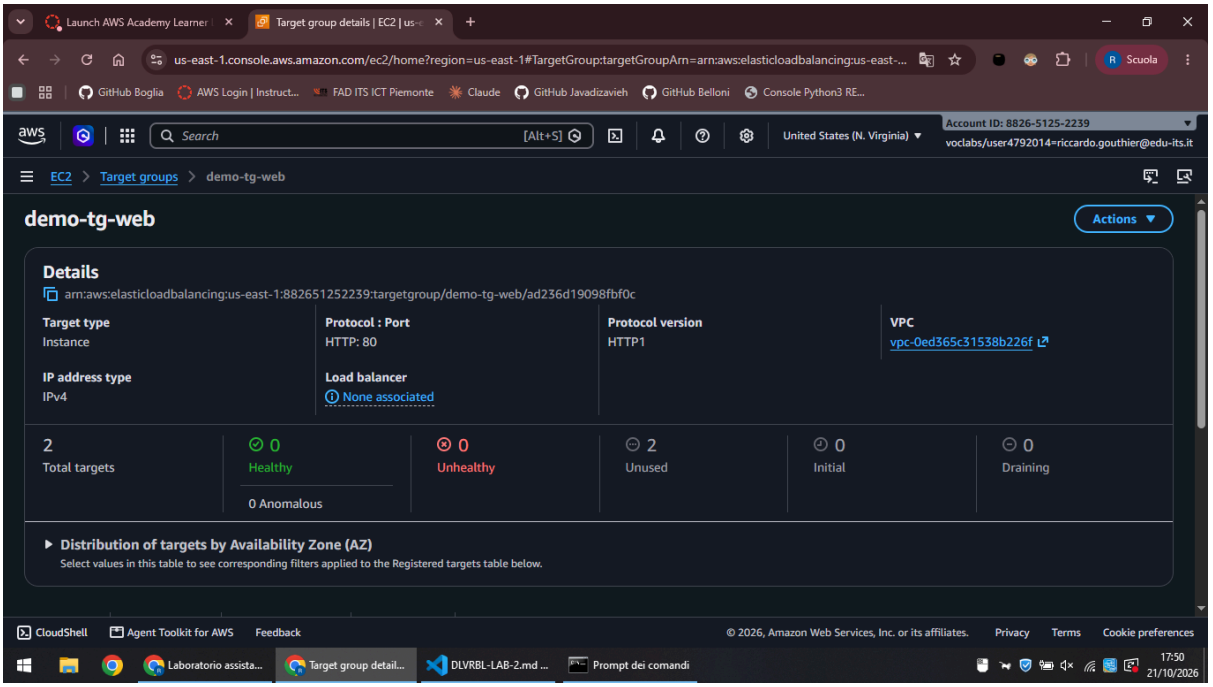


diversa AZ,

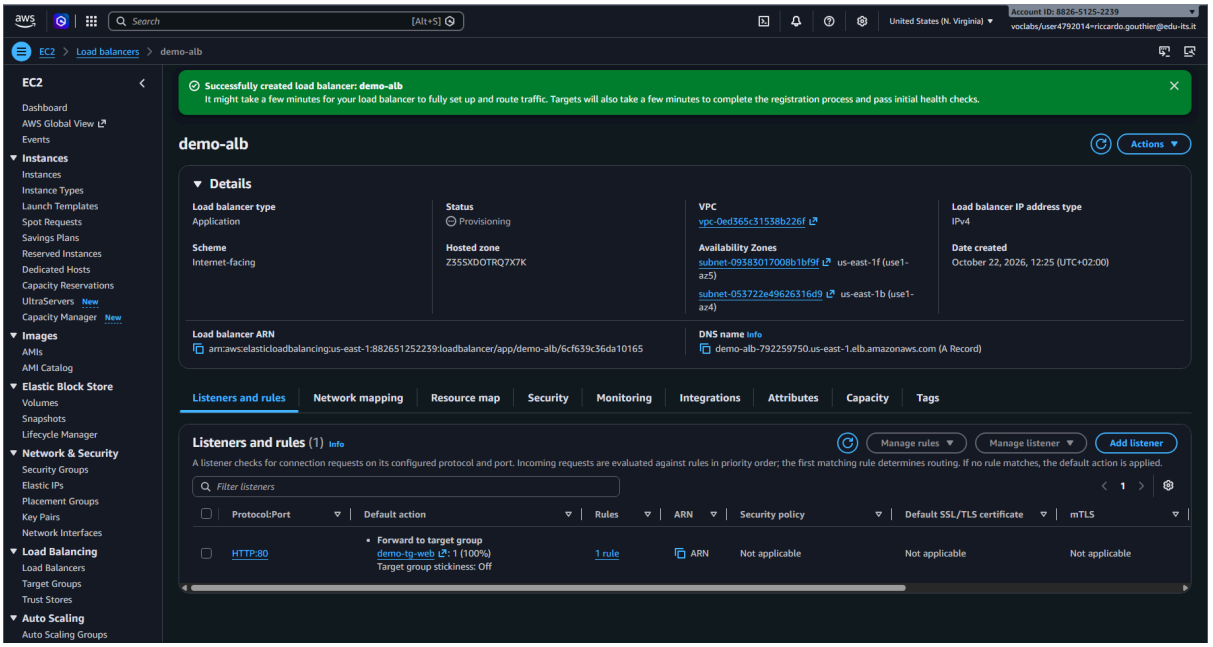
01 su 1b sg-0b56428e0de750ff7 subnet-053722e49626316d9

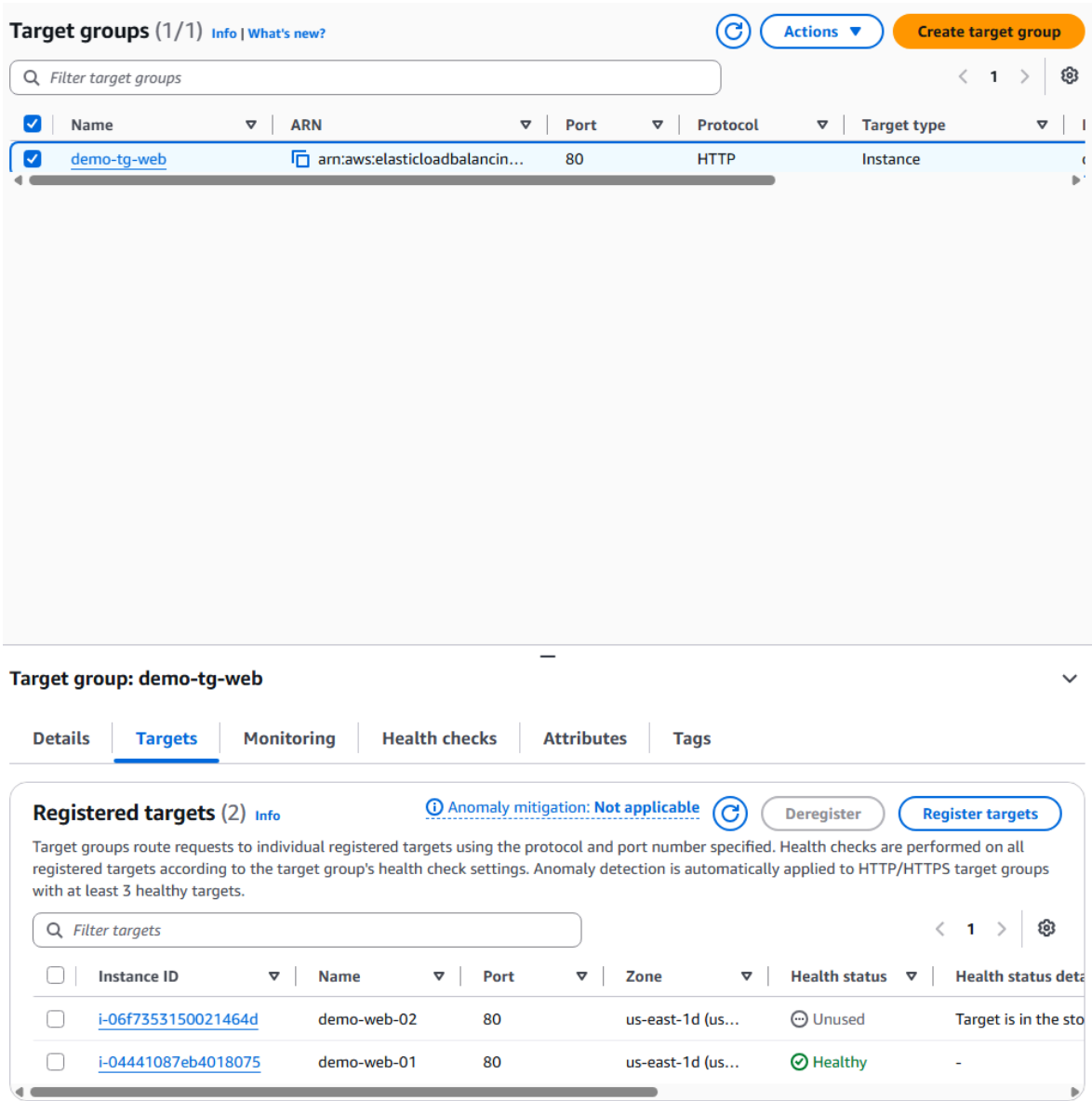
02 su 1f sg-00533f3bbbd42f395 subnet-09383017008b1bf9f

Target group

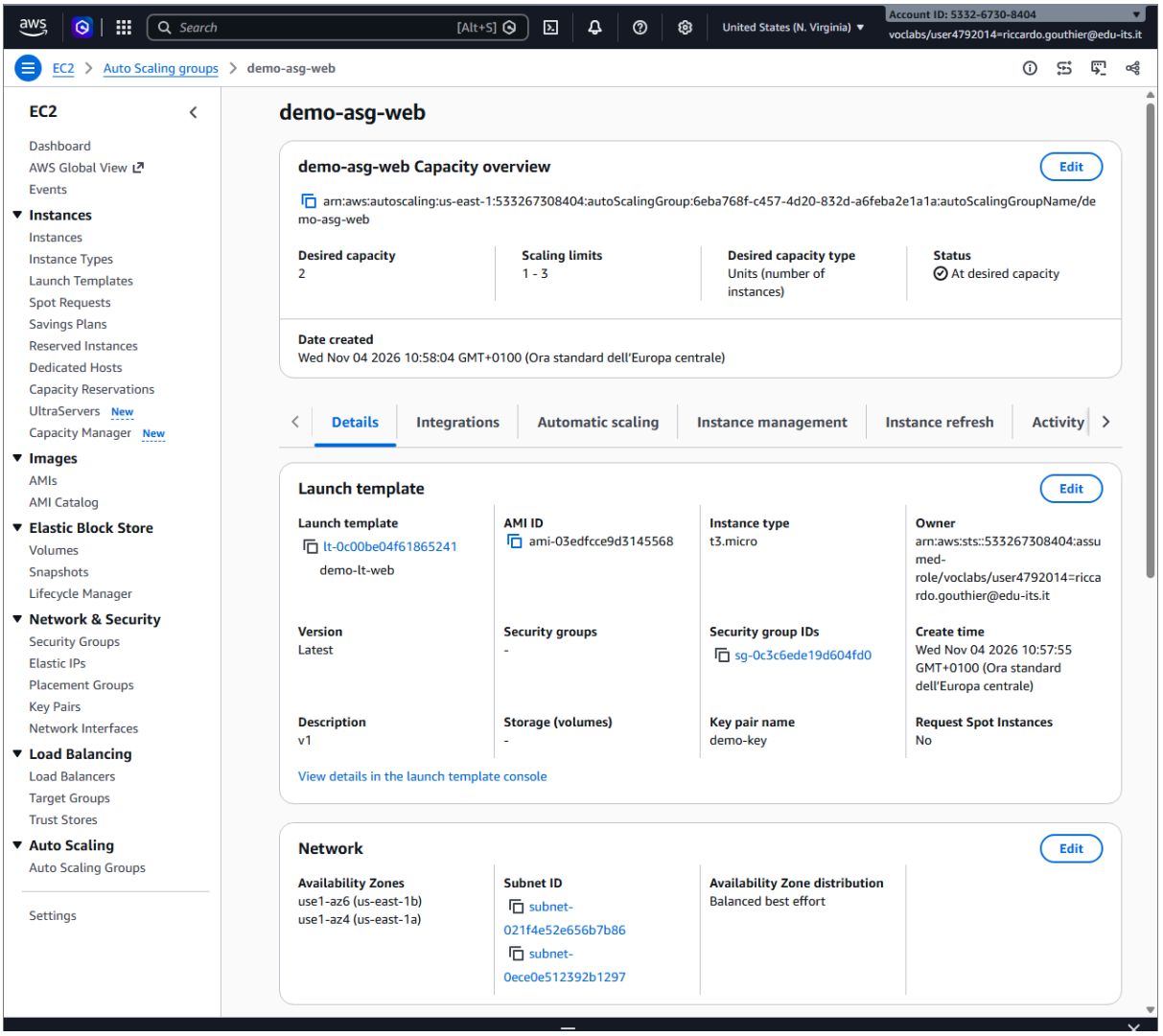


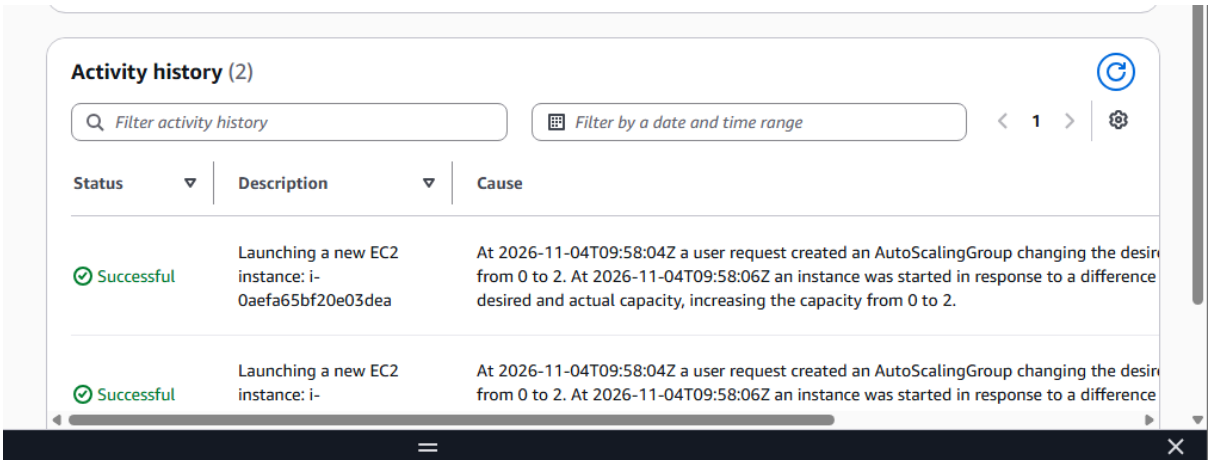
ALB

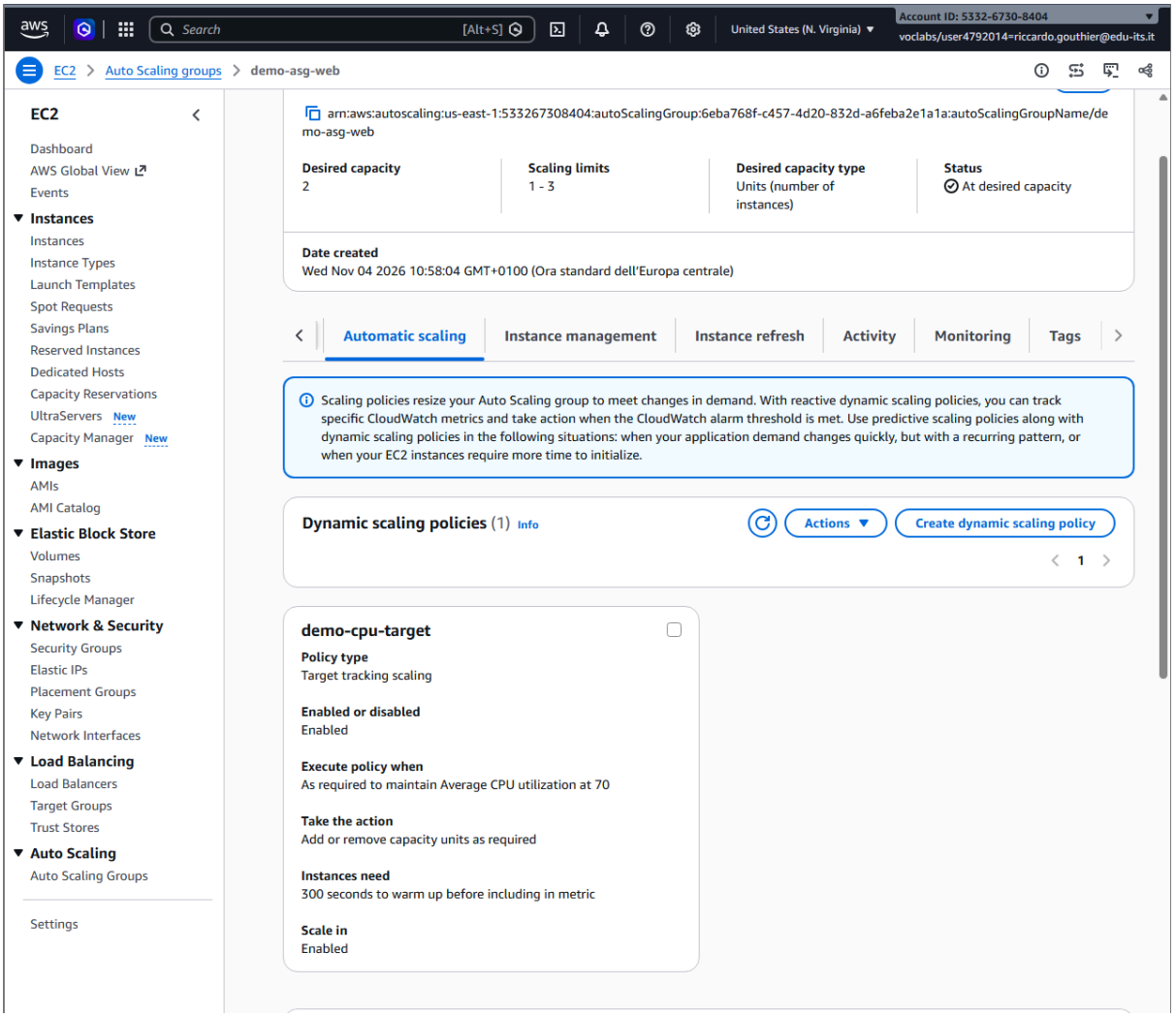

Search
[Alt+S]
United States (N. Virginia)
Account ID: 8826-5125-2239
voclabs/user4792014=riccardo.gouthier@edu-its.it
EC2 > Load balancers > demo-alb
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Successfully created load balancer: demo-alb
It might take a few minutes for your load balancer to fully set up and route traffic. Targets will also take a few minutes to complete the registration process and pass initial health checks.
demo-alb
Actions
Details
Load balancer type
Application
Status
Provisioning
VPC
vpc-0ed365c31538b226f
Load balancer IP address type
IPv4
Scheme
Internet-facing
Hosted zone
Z35SXDOTRQ7X7K
Availability Zones
subnet-09383017008b1bf9f us-east-1f (use1-az5)
subnet-053722e49626316d9 us-east-1b (use1-az4)
Date created
October 22, 2026, 12:25 (UTC+02:00)
Load balancer ARN
arn:aws:elasticloadbalancing:us-east-1:882651252239:loadbalancer/app/demo-alb/6cf639c36da10165
DNS name Info
demo-alb-792259750.us-east-1.elb.amazonaws.com (A Record)
Listeners and rules
Network mapping
Resource map
Security
Monitoring
Integrations
Attributes
Capacity
Tags
Listeners and rules (1) Info
Manage rules
Manage listener
Add listener
A listener checks for connection requests on its configured protocol and port. Incoming requests are evaluated against rules in priority order; the first matching rule determines routing. If no rule matches, the default action is applied.
Filter listeners
Protocol:Port
Default action
Rules
ARN
Security policy
Default SSL/TLS certificate
mTLS
HTTP:80
Forward to target group
demo-tg-web: 1 (100%)
Target group stickiness: Off
1 rule
ARN
Not applicable
Not applicable
Not applicable

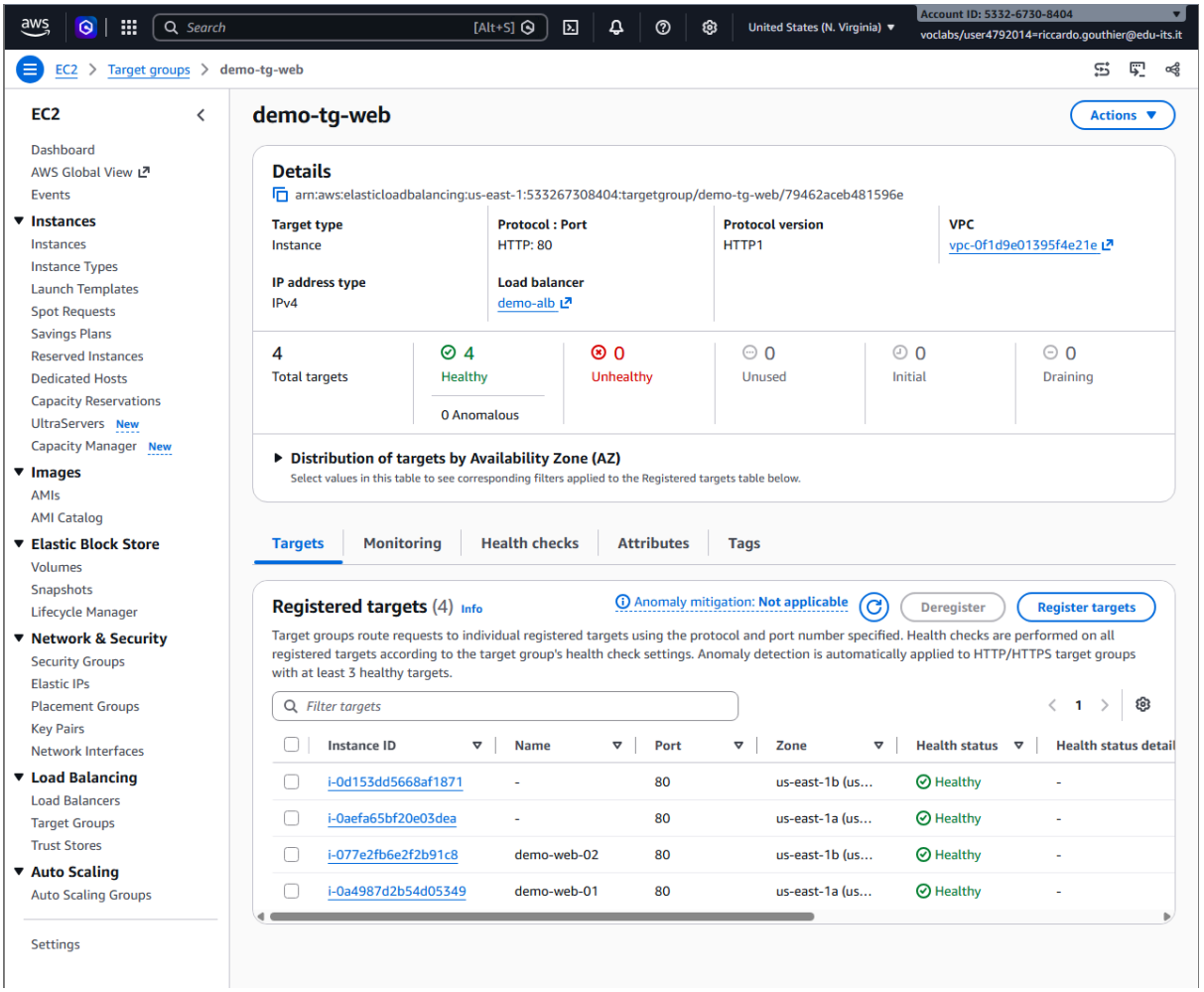

Target groups (1/1) Info | What's new?
Actions
Create target group
Filter target groups
1
Name
ARN
Port
Protocol
Target type
demo-tg-web
arn:aws:elasticloadbalancin...
80
HTTP
Instance
Target group: demo-tg-web
Details
Targets
Monitoring
Health checks
Attributes
Tags
Registered targets (2) Info
Anomaly mitigation: Not applicable
Deregister
Register targets
Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.
Filter targets
1
Instance ID
Name
Port
Zone
Health status
Health status deta
i-06f7353150021464d
demo-web-02
80
us-east-1d (us...
Unused
Target is in the sto
i-04441087eb4018075
demo-web-01
80
us-east-1d (us...
Healthy
-

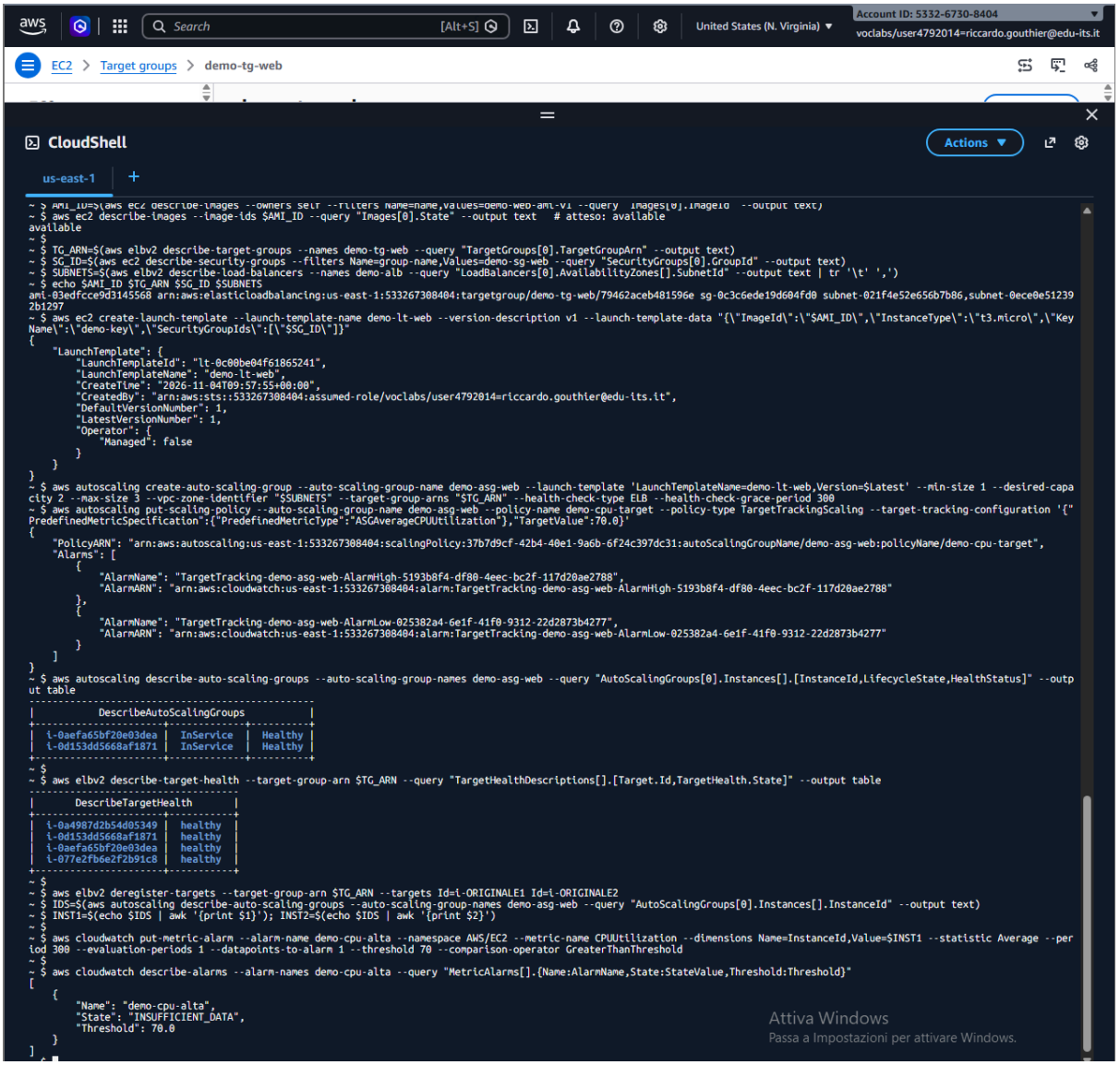
Lab 3


aws
Search
[Alt+S]
United States (N. Virginia)
Account ID: 5332-6730-8404
voclabs/user4792014=riccardo.gouthier@edu-its.it
EC2 > Auto Scaling groups > demo-asg-web
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
demo-asg-web
demo-asg-web Capacity overview
Edit
arn:aws:autoscaling:us-east-1:533267308404:autoScalingGroup:6eba768f-c457-4d20-832d-a6feba2e1a1a:autoScalingGroupName/demo-asg-web
Desired capacity
2
Scaling limits
1 - 3
Desired capacity type
Units (number of instances)
Status
At desired capacity
Date created
Wed Nov 04 2026 10:58:04 GMT+0100 (Ora standard dell'Europa centrale)
Details
Integrations
Automatic scaling
Instance management
Instance refresh
Activity
Launch template
Edit
Launch template
lt-0c00be04f61865241
demo-lt-web
AMI ID
ami-03edfcce9d3145568
Instance type
t3.micro
Owner
arn:aws:sts::533267308404:assumed-role/voclabs/user4792014=riccardo.gouthier@edu-its.it
Version
Latest
Security groups
-
Security group IDs
sg-0c3c6ede19d604fd0
Create time
Wed Nov 04 2026 10:57:55 GMT+0100 (Ora standard dell'Europa centrale)
Description
v1
Storage (volumes)
-
Key pair name
demo-key
Request Spot Instances
No
View details in the launch template console
Network
Edit
Availability Zones
use1-az6 (us-east-1b)
use1-az4 (us-east-1a)
Subnet ID
subnet-021f4e52e656b7b86
subnet-0ece0e512392b1297
Availability Zone distribution
Balanced best effort

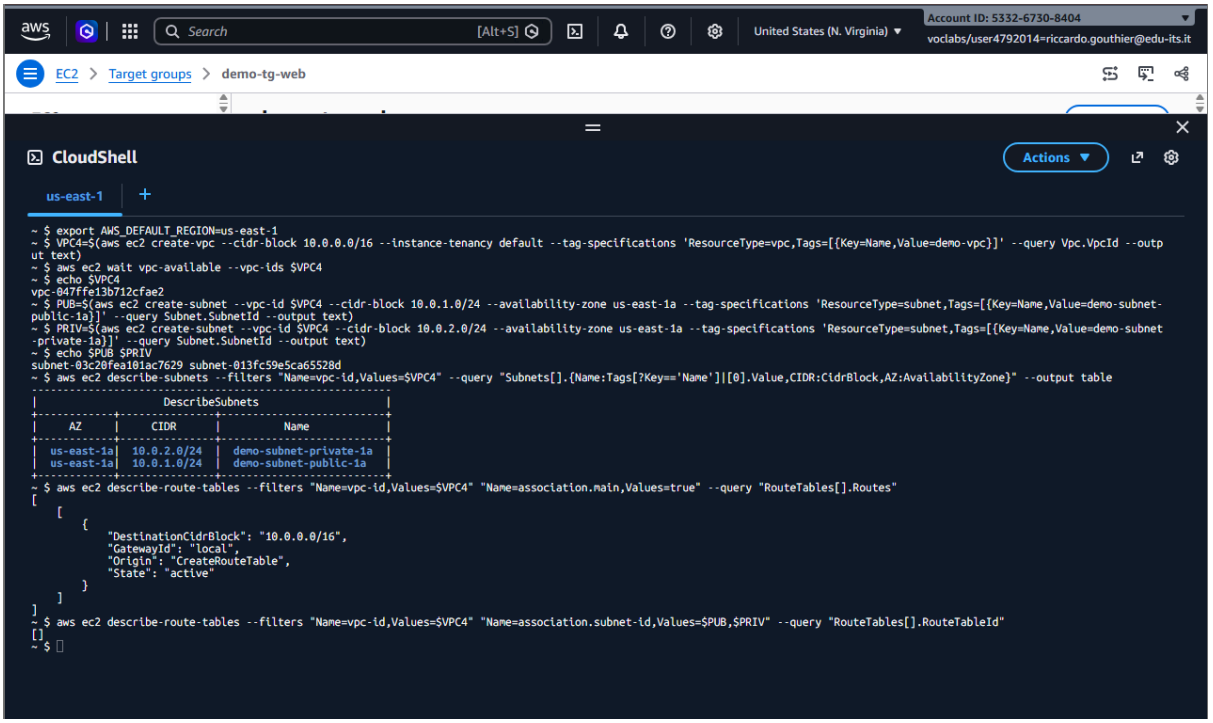

Activity history (2)
Filter activity history
Filter by a date and time range
1
Status
Description
Cause
Successful
Launching a new EC2 instance: i-0aefa65bf20e03dea
At 2026-11-04T09:58:04Z a user request created an AutoScalingGroup changing the desir from 0 to 2. At 2026-11-04T09:58:06Z an instance was started in response to a difference desired and actual capacity, increasing the capacity from 0 to 2.
Successful
Launching a new EC2 instance: i-
At 2026-11-04T09:58:04Z a user request created an AutoScalingGroup changing the desir from 0 to 2. At 2026-11-04T09:58:06Z an instance was started in response to a difference


Search
[Alt+S]
United States (N. Virginia)
Account ID: 5332-6730-8404
voclabs/user4792014=riccardo.gouthier@edu-its.it
EC2
Auto Scaling groups
demo-asg-web
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
arn:aws:autoscaling:us-east-1:533267308404:autoScalingGroup:6eba768f-c457-4d20-832d-a6feba2e1a1a:autoScalingGroupName/demo-asg-web
Desired capacity
2
Scaling limits
1 - 3
Desired capacity type
Units (number of instances)
Status
At desired capacity
Date created
Wed Nov 04 2026 10:58:04 GMT+0100 (Ora standard dell'Europa centrale)
Automatic scaling
Instance management
Instance refresh
Activity
Monitoring
Tags
Scaling policies resize your Auto Scaling group to meet changes in demand. With reactive dynamic scaling policies, you can track specific CloudWatch metrics and take action when the CloudWatch alarm threshold is met. Use predictive scaling policies along with dynamic scaling policies in the following situations: when your application demand changes quickly, but with a recurring pattern, or when your EC2 instances require more time to initialize.
Dynamic scaling policies (1) Info
Actions
Create dynamic scaling policy
1
demo-cpu-target
Policy type
Target tracking scaling
Enabled or disabled
Enabled
Execute policy when
As required to maintain Average CPU utilization at 70
Take the action
Add or remove capacity units as required
Instances need
300 seconds to warm up before including in metric
Scale in
Enabled


Search
[Alt+S]
United States (N. Virginia)
Account ID: 5332-6730-8404
voclabs/user4792014=riccardo.gouthier@edu-its.it
EC2 > Target groups > demo-tg-web
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
UltraServers New
Capacity Manager New
Images
AMIs
AMI Catalog
Elastic Block Store
Volumes
Snapshots
Lifecycle Manager
Network & Security
Security Groups
Elastic IPs
Placement Groups
Key Pairs
Network Interfaces
Load Balancing
Load Balancers
Target Groups
Trust Stores
Auto Scaling
Auto Scaling Groups
Settings
demo-tg-web
Actions
Details
arn:aws:elasticloadbalancing:us-east-1:533267308404:targetgroup/demo-tg-web/79462aceb481596e
Target type
Instance
Protocol : Port
HTTP: 80
Protocol version
HTTP1
VPC
vpc-0f1d9e01395f4e21e
IP address type
IPv4
Load balancer
demo-alb
4
Total targets
4
Healthy
0 Anomalous
0
Unhealthy
0
Unused
0
Initial
0
Draining
Distribution of targets by Availability Zone (AZ)
Select values in this table to see corresponding filters applied to the Registered targets table below.
Targets
Monitoring
Health checks
Attributes
Tags
Registered targets (4) Info
Anomaly mitigation: Not applicable
Deregister
Register targets
Target groups route requests to individual registered targets using the protocol and port number specified. Health checks are performed on all registered targets according to the target group's health check settings. Anomaly detection is automatically applied to HTTP/HTTPS target groups with at least 3 healthy targets.
Filter targets
1
Instance ID
Name
Port
Zone
Health status
Health status detail
i-0d153dd5668af1871 - 80 us-east-1b (us... Healthy -
i-0aefa65bf20e03dea - 80 us-east-1a (us... Healthy -
i-077e2fb6e2f2b91c8 demo-web-02 80 us-east-1b (us... Healthy -
i-0a4987d2b54d05349 demo-web-01 80 us-east-1a (us... Healthy -


Search
[Alt+S]
United States (N. Virginia)
Account ID: 5332-6730-8404
voclabs/user4792014=riccardo.gouthier@edu-its.it
EC2 > Target groups > demo-tg-web
CloudShell
Actions
us-east-1
~ $ AMI_ID=$(aws ec2 describe-images --owners self --filters Name=name,Values=demo-web-ami-v1 --query 'Images[0].ImageId' --output text)
~ $ aws ec2 describe-images --image-ids $AMI_ID --query "Images[0].State" --output text # atteso: available
available
~ $
~ $ TG_ARN=$(aws elbv2 describe-target-groups --names demo-tg-web --query "TargetGroups[0].TargetGroupArn" --output text)
~ $ SG_ID=$(aws ec2 describe-security-groups --filters Name=group-name,Values=demo-sg-web --query "SecurityGroups[0].GroupId" --output text)
~ $ SUBNETS=$(aws elbv2 describe-load-balancers --names demo-alb --query "LoadBalancers[0].AvailabilityZones[].SubnetId" --output text | tr '\t' ',')
~ $ echo $AMI_ID $TG_ARN $SG_ID $SUBNETS
ami-03edfcce9d3145568 arn:aws:elasticloadbalancing:us-east-1:533267308404:targetgroup/demo-tg-web/79462aceb481596e sg-0c3c6ede19d604fd0 subnet-021f4e52e656b7b86,subnet-0ece0e512392b1297
~ $ aws ec2 create-launch-template --launch-template-name demo-lt-web --version-description v1 --launch-template-data "{\"ImageId\":\"$AMI_ID\",\"InstanceType\":\"t3.micro\",\"KeyName\":\"demo-key\",\"SecurityGroupIds\":[\"$SG_ID\"]}"
{
"LaunchTemplate": {
"LaunchTemplateId": "lt-0c00be04f61865241",
"LaunchTemplateName": "demo-lt-web",
"CreateTime": "2026-11-04T09:57:55+00:00",
"CreatedBy": "arn:aws:sts::533267308404:assumed-role/voclabs/user4792014=riccardo.gouthier@edu-its.it",
"DefaultVersionNumber": 1,
"LatestVersionNumber": 1,
"Operator": {
"Managed": false
}
}
}
~ $ aws autoscaling create-auto-scaling-group --auto-scaling-group-name demo-asg-web --launch-template 'LaunchTemplateName=demo-lt-web,Version=$Latest' --min-size 1 --desired-capacity 2 --max-size 3 --vpc-zone-identifier "$SUBNETS" --target-group-arns "$TG_ARN" --health-check-type ELB --health-check-grace-period 300
~ $ aws autoscaling put-scaling-policy --auto-scaling-group-name demo-asg-web --policy-name demo-cpu-target --policy-type TargetTrackingScaling --target-tracking-configuration '{"PredefinedMetricSpecification":{"PredefinedMetricType":"ASGAverageCPUUtilization"},"TargetValue":70.0}'
{
"PolicyARN": "arn:aws:autoscaling:us-east-1:533267308404:scalingPolicy:37b7d9cf-42b4-40e1-9a6b-6f24c397dc31:autoScalingGroupName/demo-asg-web:policyName/demo-cpu-target",
"Alarms": [
{
"AlarmName": "TargetTracking-demo-asg-web-AlarmHigh-5193b8f4-df80-4eec-bc2f-117d20ae2788",
"AlarmARN": "arn:aws:cloudwatch:us-east-1:533267308404:alarm:TargetTracking-demo-asg-web-AlarmHigh-5193b8f4-df80-4eec-bc2f-117d20ae2788"
},
{
"AlarmName": "TargetTracking-demo-asg-web-AlarmLow-025382a4-6e1f-41f0-9312-22d2873b4277",
"AlarmARN": "arn:aws:cloudwatch:us-east-1:533267308404:alarm:TargetTracking-demo-asg-web-AlarmLow-025382a4-6e1f-41f0-9312-22d2873b4277"
}
]
}
~ $ aws autoscaling describe-auto-scaling-groups --auto-scaling-group-names demo-asg-web --query "AutoScalingGroups[0].Instances[].[InstanceId,LifecycleState,HealthStatus]" --output table
DescribeAutoScalingGroups
i-0aefa65bf20e03dea | InService | Healthy
i-0d153dd5668af1871 | InService | Healthy
~ $
~ $ aws elbv2 describe-target-health --target-group-arn $TG_ARN --query "TargetHealthDescriptions[].[Target.Id,TargetHealth.State]" --output table
DescribeTargetHealth
i-0a4987d2b54d05349 | healthy
i-0d153dd5668af1871 | healthy
i-0aefa65bf20e03dea | healthy
i-077e2fb6e2f2b91c8 | healthy
~ $
~ $ aws elbv2 deregister-targets --target-group-arn $TG_ARN --targets Id=i-ORIGINALE1 Id=i-ORIGINALE2
~ $ IDS=$(aws autoscaling describe-auto-scaling-groups --auto-scaling-group-names demo-asg-web --query "AutoScalingGroups[0].Instances[].InstanceId" --output text)
~ $ INST1=$(echo $IDS | awk '{print $1}'); INST2=$(echo $IDS | awk '{print $2}')
~ $
~ $ aws cloudwatch put-metric-alarm --alarm-name demo-cpu-alta --namespace AWS/EC2 --metric-name CPUUtilization --dimensions Name=InstanceId,Value=$INST1 --statistic Average --period 300 --evaluation-periods 1 --datapoints-to-alarm 1 --threshold 70 --comparison-operator GreaterThanThreshold
~ $
~ $ aws cloudwatch describe-alarms --alarm-names demo-cpu-alta --query "MetricAlarms[].{Name:AlarmName,State:StateValue,Threshold:Threshold}"
[
{
"Name": "demo-cpu-alta",
"State": "INSUFFICIENT_DATA",
"Threshold": 70.0
}
]
Attiva Windows
Passa a Impostazioni per attivare Windows.

Lab 4



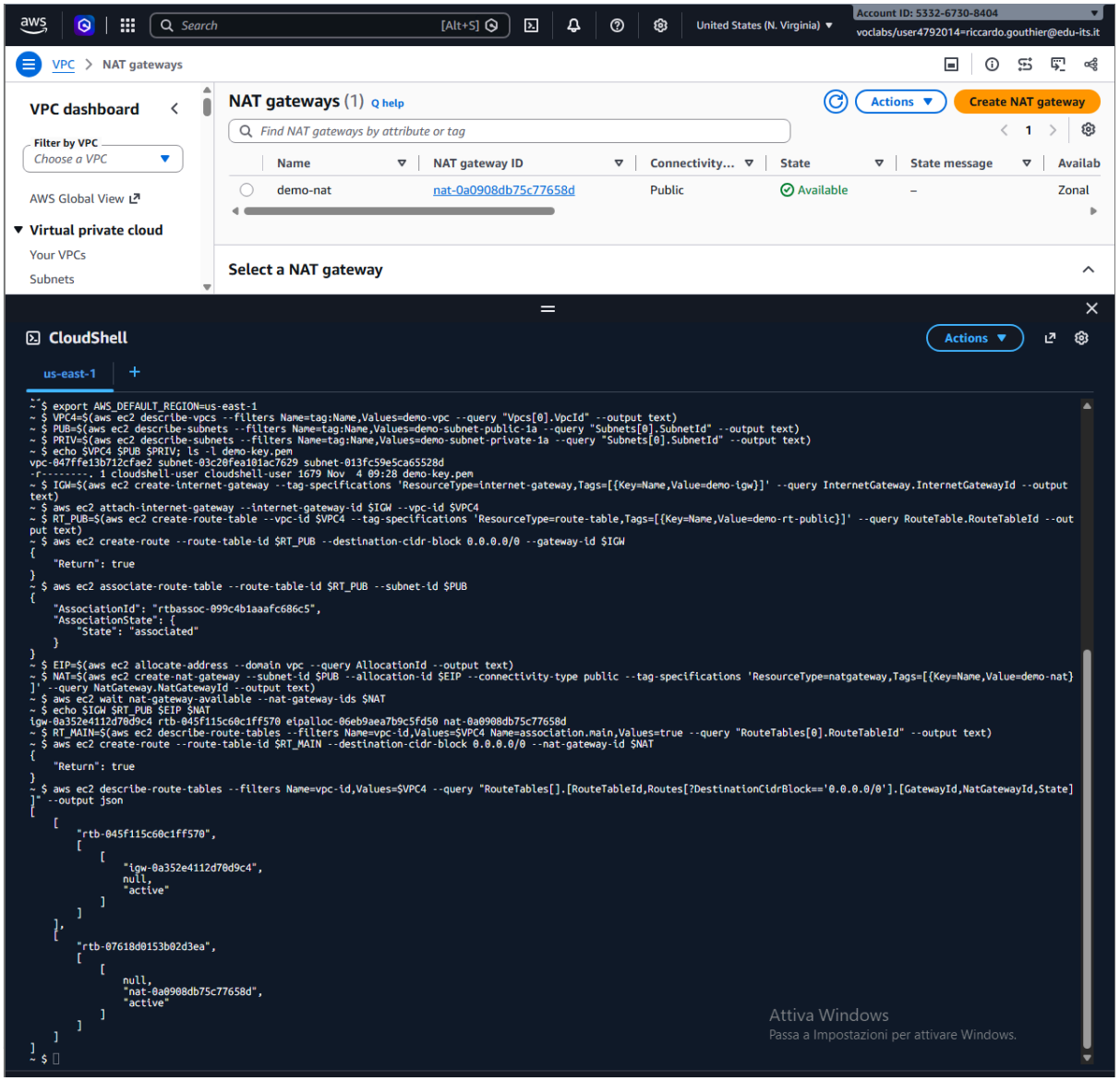
vpc separate

Nome	CIDR	AZ	Tipo
demo-vpc	10.0.0.0/16	—	Rete dedicata
demo-subnet-public-1a	10.0.1.0/24	us-east-1a	pubblica (pianificata)
demo-subnet-private-1a	10.0.2.0/24	us-east-1a	privata

Nessuna subnet realmente pubblica: manca rotta 0.0.0.0/0 → IGW. public = solo nome.

Lab 5

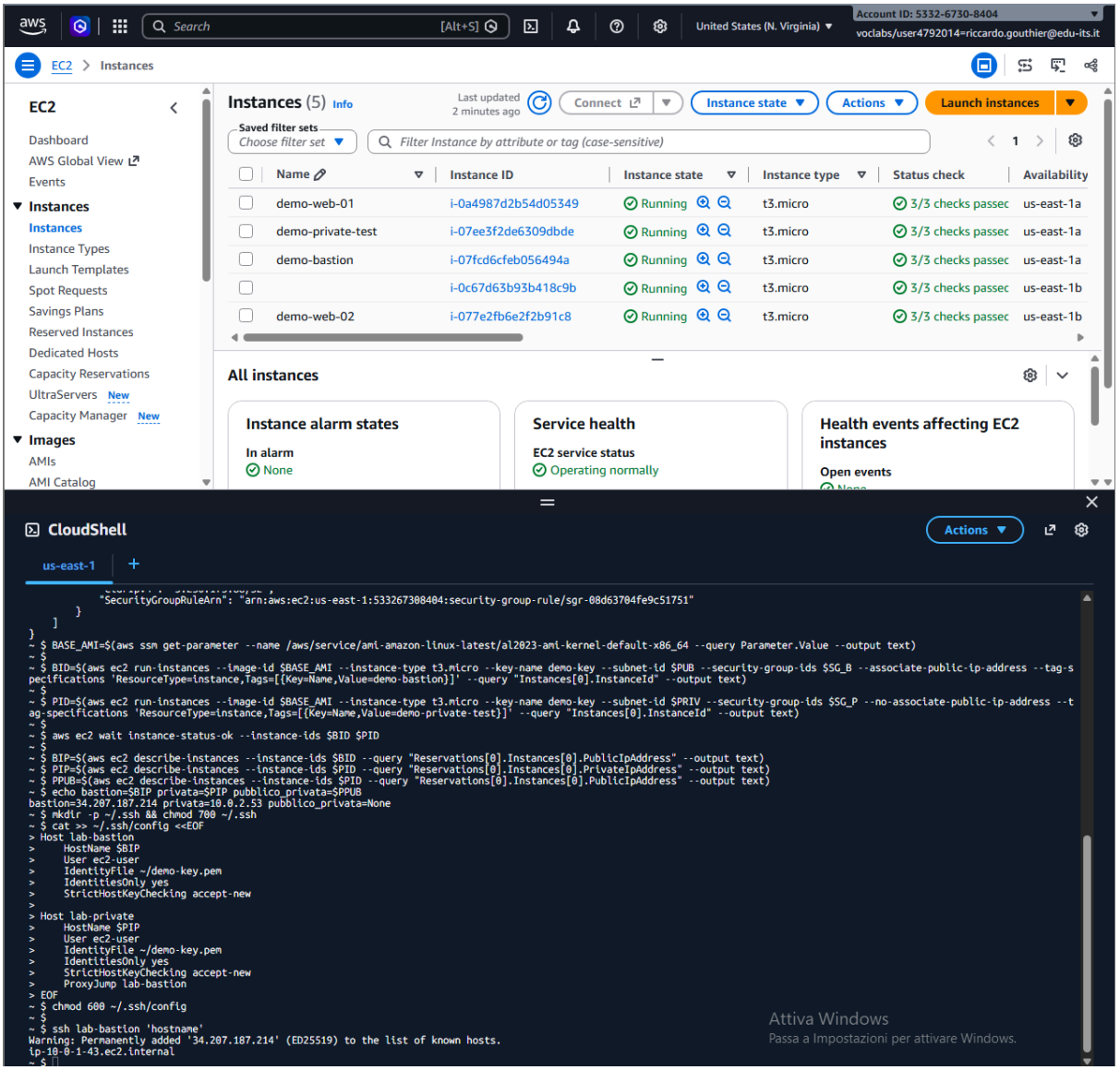
Variabili, Internal Gateway, demo-rt-public + associazione, Elastic IP + NAT demo-nat,Rotta privata → NAT (main route table) + verifica



Security Group

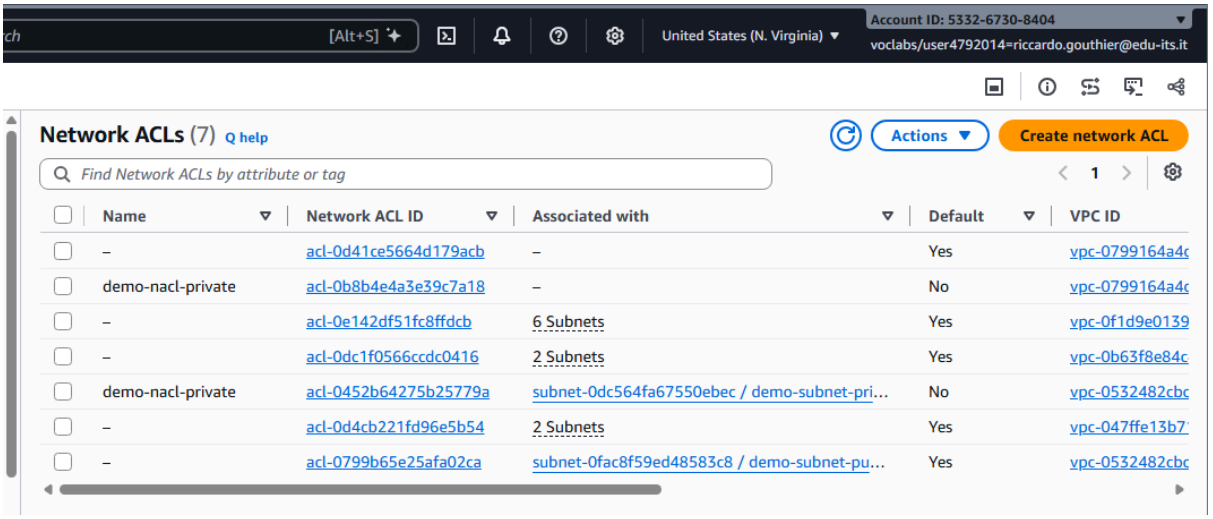


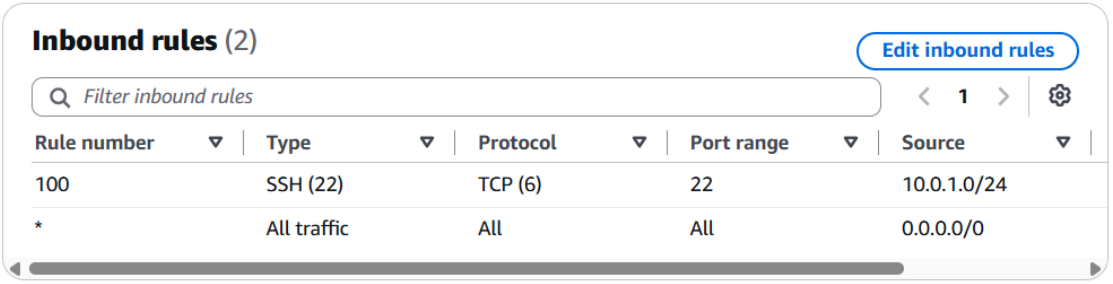
Istanze, configurazione SSH

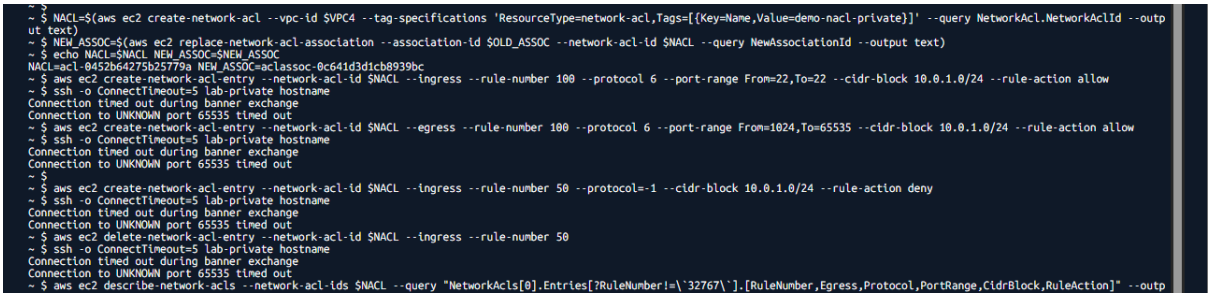


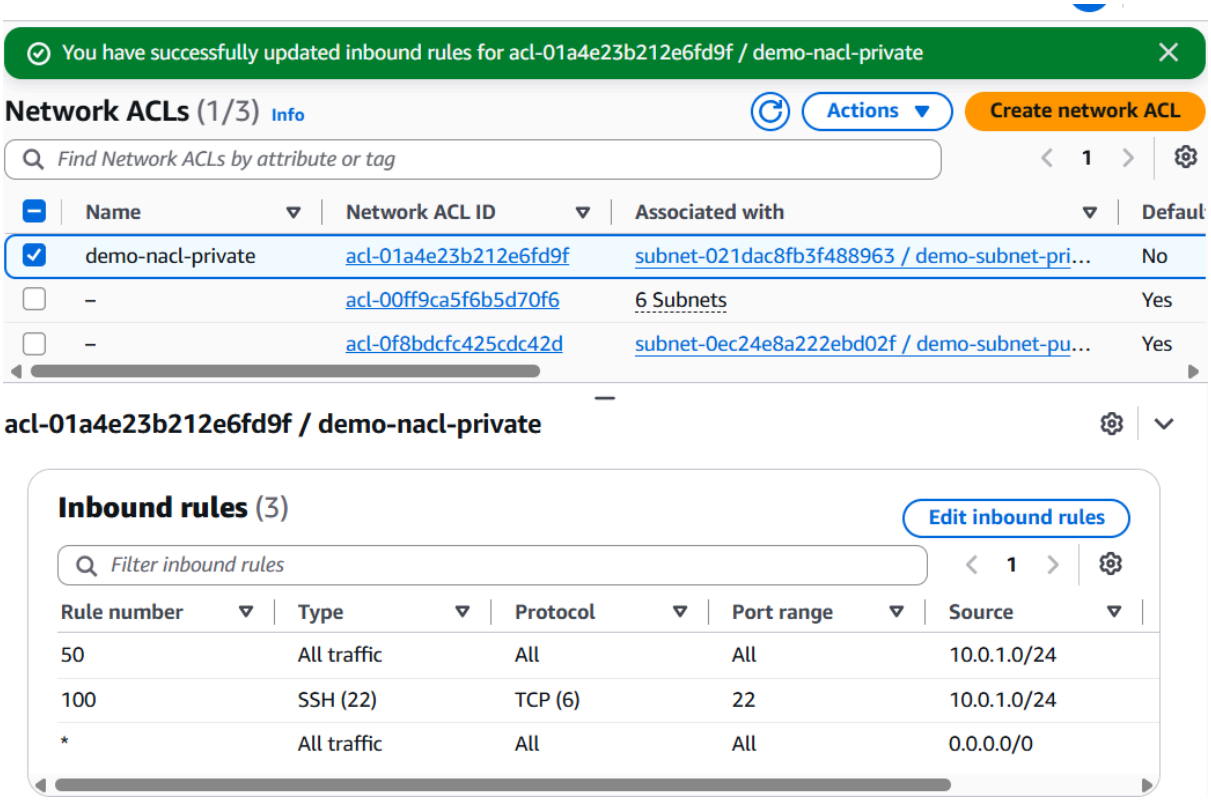
Rotte: pubblica 0.0.0.0/0 → demo-igw; privata 0.0.0.0/0 → demo-nat
demo-sg-bastion: 22 da $MYIP/32; demo-sg-private: 22 da $SG_B
Salto SSH + curl annotati

Lab 6


[Alt+S]
United States (N. Virginia)
Account ID: 5332-6730-8404
voclabs/user4792014=riccardo.gouthier@edu-its.it
Network ACLs (7) Q help
Actions
Create network ACL
Find Network ACLs by attribute or tag
1
Name
Network ACL ID
Associated with
Default
VPC ID
– acl-0d41ce5664d179acb – Yes vpc-0799164a4c
demo-nacl-private acl-0b8b4e4a3e39c7a18 – No vpc-0799164a4c
– acl-0e142df51fc8ffdcb 6 Subnets Yes vpc-0f1d9e0139
– acl-0dc1f0566ccdc0416 2 Subnets Yes vpc-0b63f8e84c
demo-nacl-private acl-0452b64275b25779a subnet-0dc564fa67550ebec / demo-subnet-pri... No vpc-0532482cbc
– acl-0d4cb221fd96e5b54 2 Subnets Yes vpc-047ffe13b7
– acl-0799b65e25afa02ca subnet-0fac8f59ed48583c8 / demo-subnet-pu... Yes vpc-0532482cbc


Inbound rules (2)
Edit inbound rules
Filter inbound rules
1
Rule number
Type
Protocol
Port range
Source
100 SSH (22) TCP (6) 22 10.0.1.0/24
* All traffic All All 0.0.0.0/0


~ $
~ $ NACL=$(aws ec2 create-network-acl --vpc-id $VPC4 --tag-specifications 'ResourceType=network-acl,Tags=[{Key=Name,Value=demo-nacl-private}]' --query NetworkAcl.NetworkAclId --output text)
~ $ NEW_ASSOC=$(aws ec2 replace-network-acl-association --association-id $OLD_ASSOC --network-acl-id $NACL --query NewAssociationId --output text)
~ $ echo NACL=$NACL NEW_ASSOC=$NEW_ASSOC
NACL=acl-0452b64275b25779a NEW_ASSOC=aclassoc-0c641d3d1cb8939bc
~ $ aws ec2 create-network-acl-entry --network-acl-id $NACL --ingress --rule-number 100 --protocol 6 --port-range From=22,To=22 --cidr-block 10.0.1.0/24 --rule-action allow
~ $ ssh -o ConnectTimeout=5 lab-private hostname
Connection timed out during banner exchange
Connection to UNKNOWN port 65535 timed out
~ $ aws ec2 create-network-acl-entry --network-acl-id $NACL --egress --rule-number 100 --protocol 6 --port-range From=1024,To=65535 --cidr-block 10.0.1.0/24 --rule-action allow
~ $ ssh -o ConnectTimeout=5 lab-private hostname
Connection timed out during banner exchange
Connection to UNKNOWN port 65535 timed out
~ $
~ $ aws ec2 create-network-acl-entry --network-acl-id $NACL --ingress --rule-number 50 --protocol=-1 --cidr-block 10.0.1.0/24 --rule-action deny
~ $ ssh -o ConnectTimeout=5 lab-private hostname
Connection timed out during banner exchange
Connection to UNKNOWN port 65535 timed out
~ $ aws ec2 delete-network-acl-entry --network-acl-id $NACL --ingress --rule-number 50
~ $ ssh -o ConnectTimeout=5 lab-private hostname
Connection timed out during banner exchange
Connection to UNKNOWN port 65535 timed out
~ $ aws ec2 describe-network-acls --network-acl-ids $NACL --query "NetworkAcls[0].Entries[?RuleNumber!=\`32767\`].[RuleNumber,Egress,Protocol,PortRange,CidrBlock,RuleAction]" --outp



Lab 7

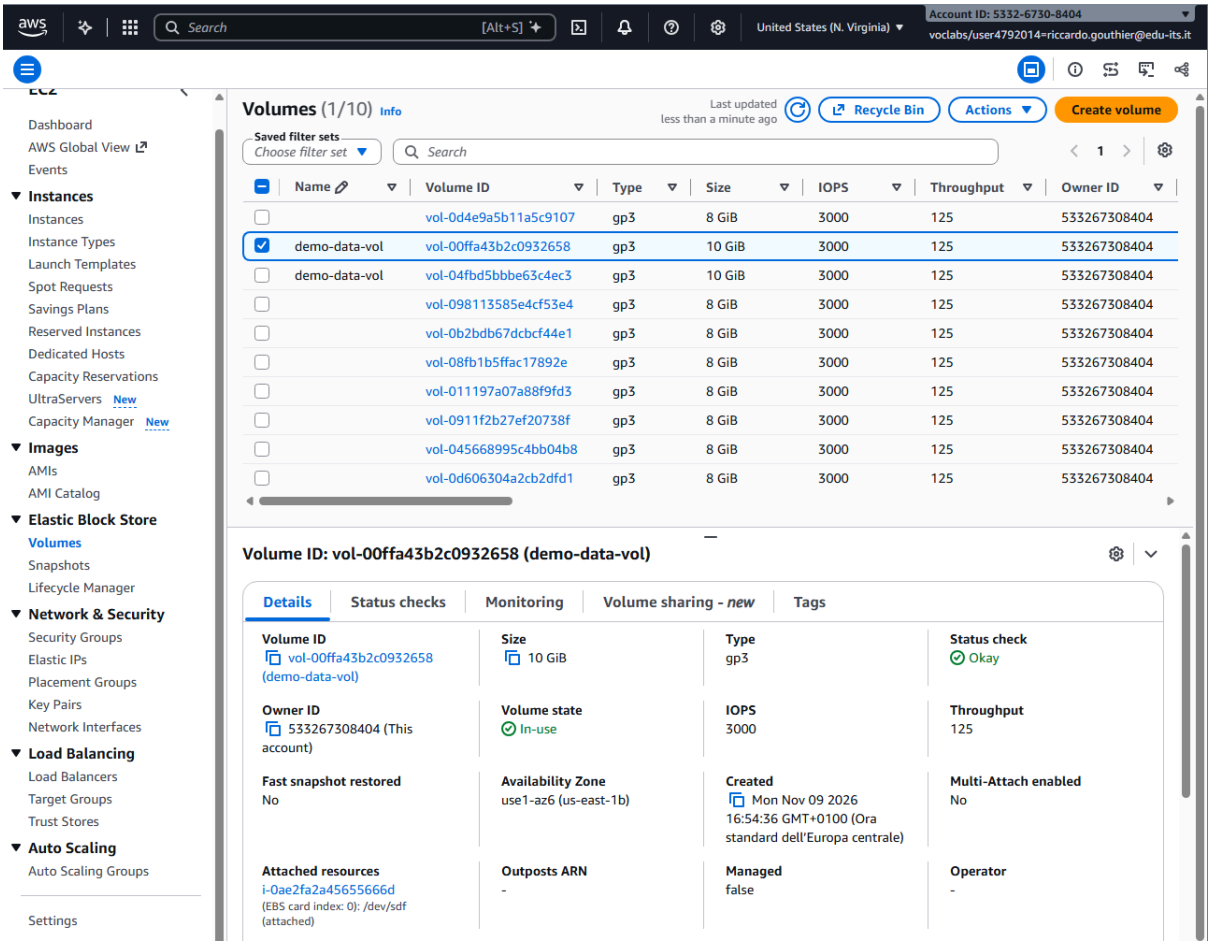
Confronto e scelta dell'istanza — passi 1–2

EBS: storage a blocchi per istanza EC2 nella stessa Availability Zone. Adatto a dischi dati. EFS: filesystem condiviso fra più istanze. S3: oggetti via API. Adatto ad archivi e backup.

Istanza Linux autonoma scelta: <nome-istanza>, <instance-id>, <tipo>, us-east-1b. Nessun Auto Scaling Group. Configurazione via CLI da CloudShell.

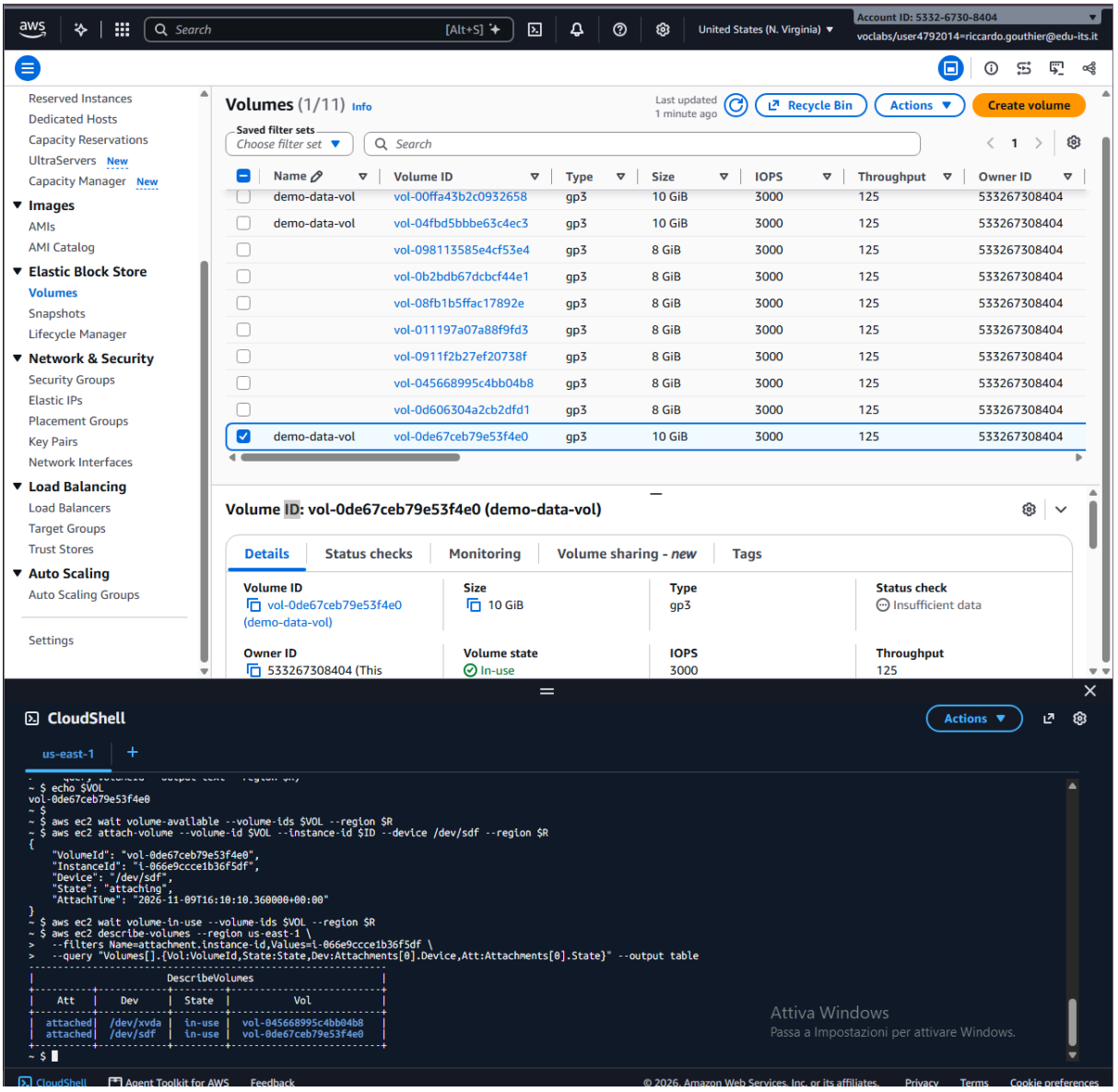
Creazione e allegamento volume — passi 3–4

AZ istanza letta via describe-instances: us-east-1b. Volume demo-data-vol: vol-00ffa43b2c0932658, gp3, 10 GiB, us-east-1b. Atteso available. Allegato su /dev/sdf con attach-volume. Stato verificato: in-use, allegato alla sola istanza del lab.



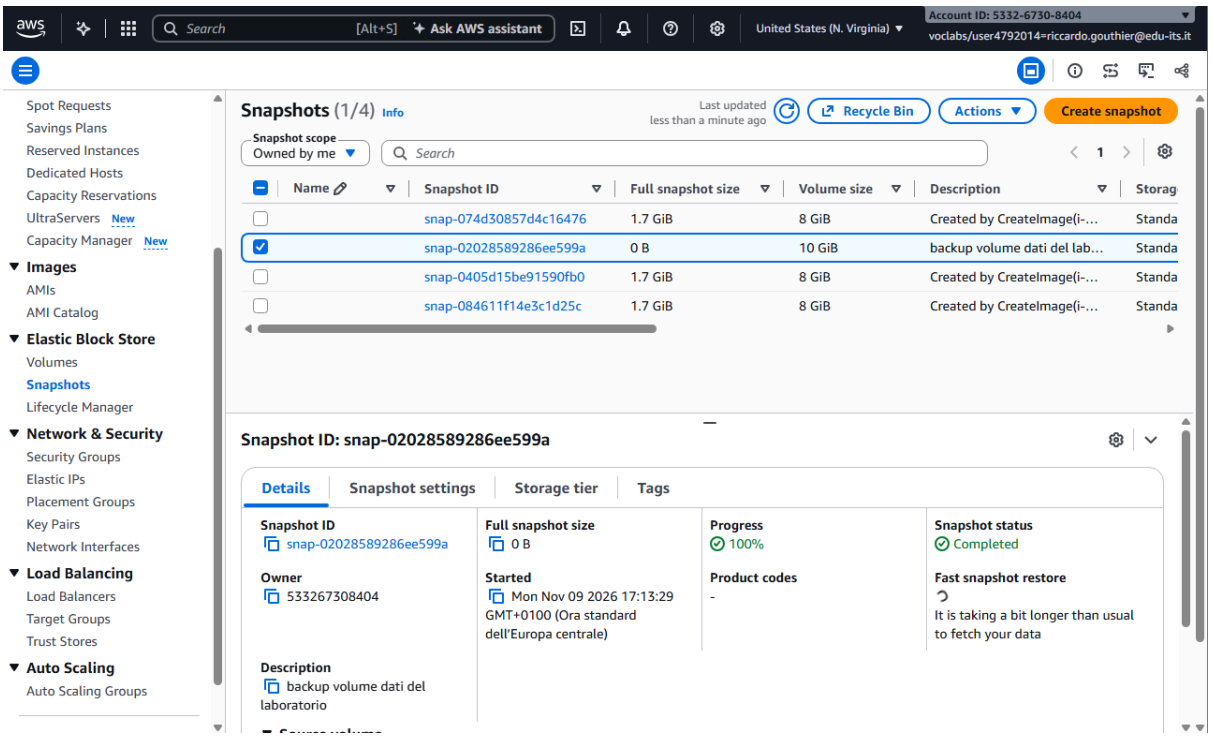
Identificazione, filesystem e prova dati — passi 5–6

SSH riuscito. lsblk -f ha mostrato nuovo disco 10 GiB senza filesystem: <xvdf / nvme1n1>. blkid senza output → disco vuoto. Formattato solo il nuovo disco in ext4, montato in /mnt/demo-data. findmnt conferma mount. test.txt scritto e riletto: «demo storage EBS».



Backup verificato — passo 7

Dopo sync, snapshot <snapshot-id>, descrizione «backup volume dati del laboratorio». Stato verificato: completed, progresso 100%.



Documentazione e conservazione — passo 8

Volume e snapshot <conservati per verifica docente e Lab 10, 12, 14 / eliminati>. Cleanup <non eseguito / eseguito>. Nessuna risorsa preesistente eliminata